Estimating Census Block-Level Citizen Voting Age Population for Redistricting

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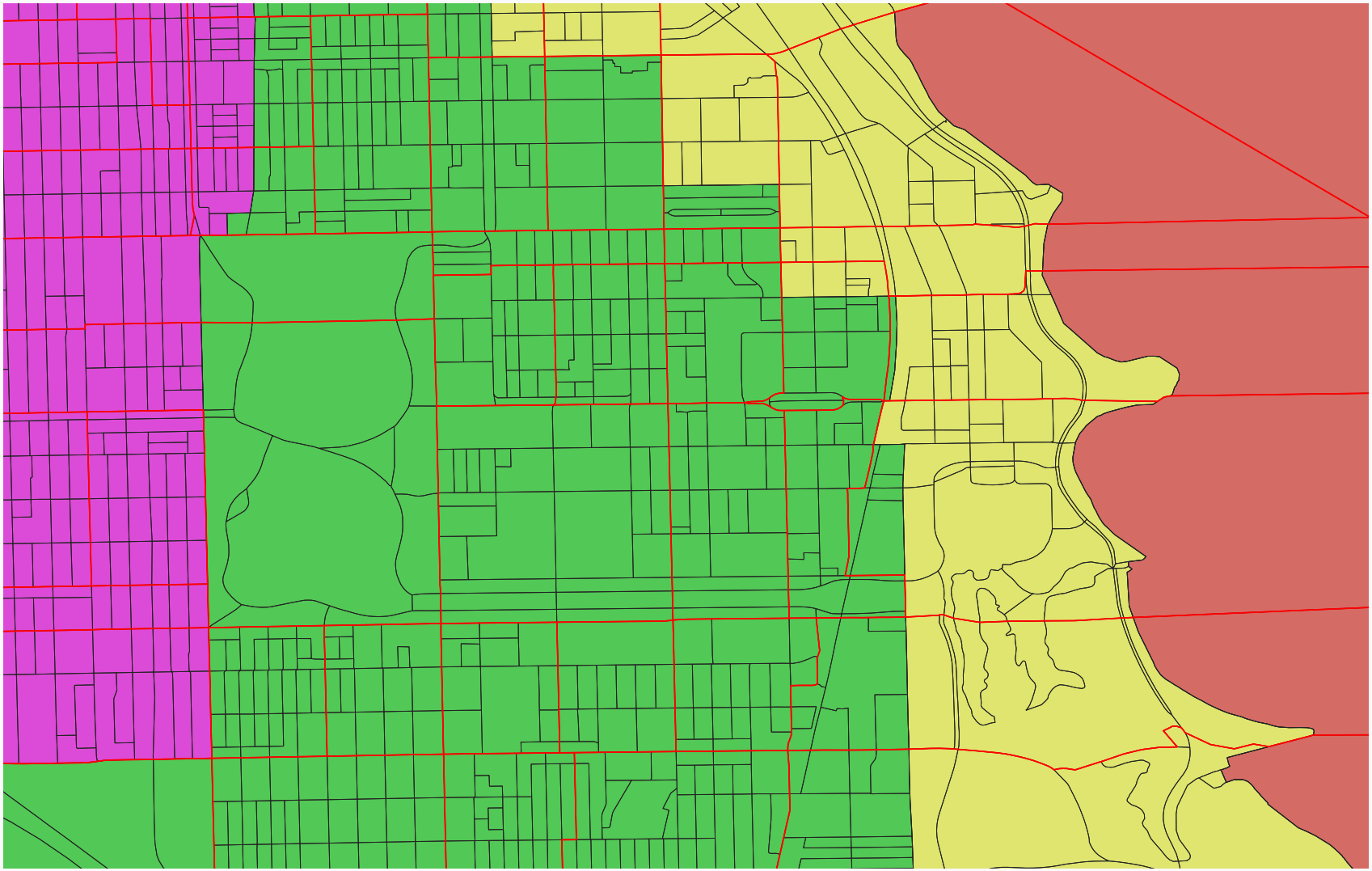
Background

Challenges

1. Citizenship data is only available on tracts, but we need it on blocks.

2. The race/ethnicity categories for which we have citizenship data are different from those relevant for litigation.

Redistricting Geography



Data Sources

- Decennial Census
  - Block-level data
  - Only gives voting age population (VAP)
  - Full race/ethnicity breakdown
- American Community Survey (ACS)
  - Tract level data
  - Both VAP and CVAP
  - Only aggregates of race/ethnicity tensor

From Tracts to Blocks

Naive Disaggregation

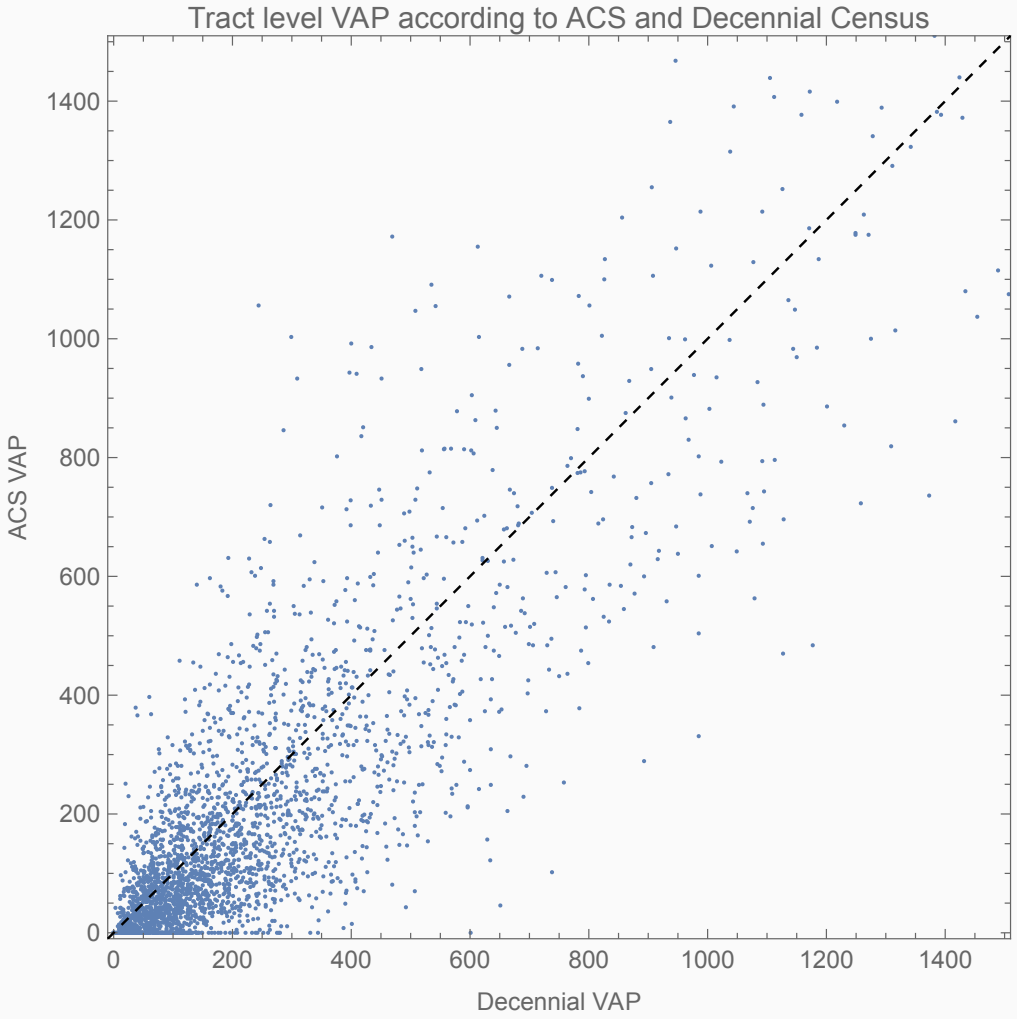
Let b be some Census Block, and t be the tract containing b.

$$CVAP_b \approx CVAP_t^{ACS} \frac{VAP_b^{dec}}{VAP_t^{dec}}$$

Or worse, where A_i gives the land area of the geography i:

$$CVAP_b \approx CVAP_t^{ACS} \frac{A_b}{A_t}$$

ACS vs. Decennial Census



Discounted VAP

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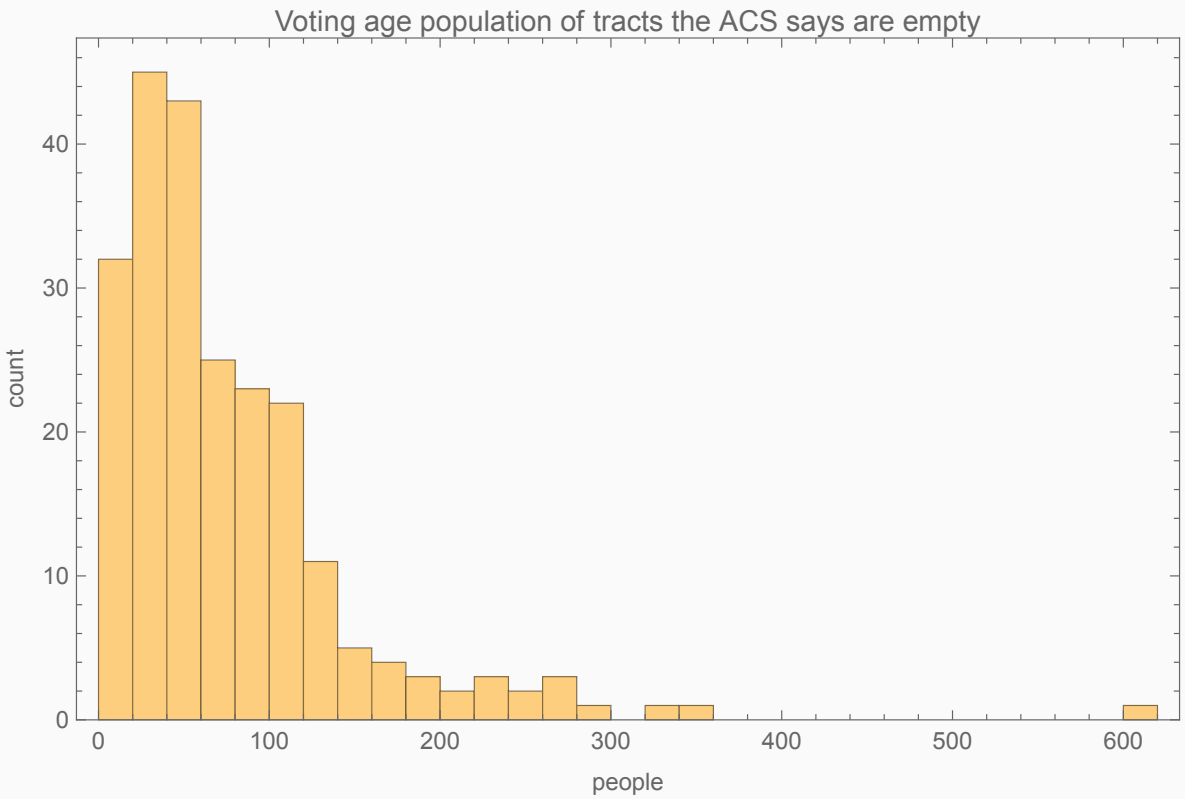
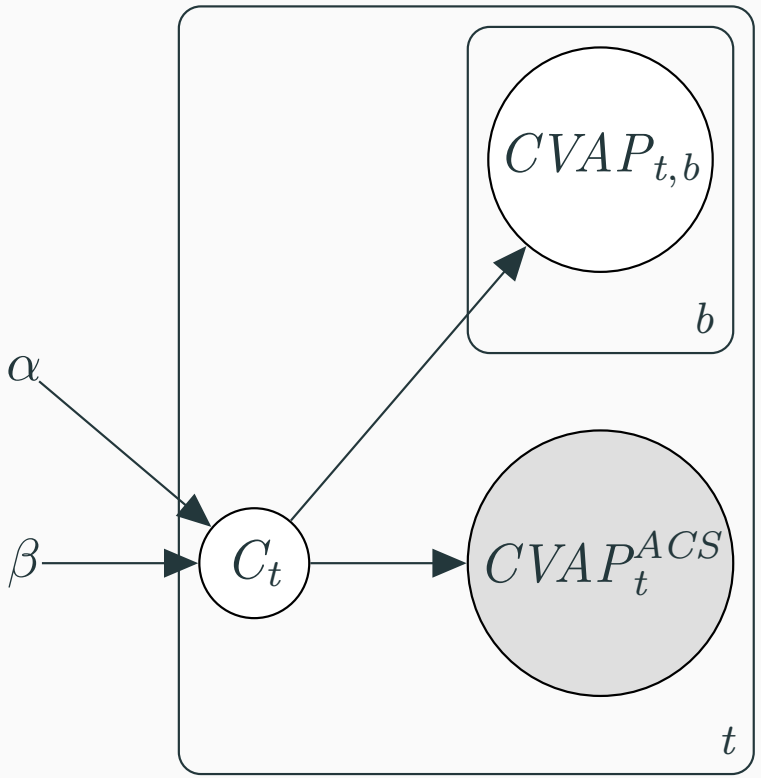


Beta-Binomial Model

For all tracts t and blocks b in t,

$$C_t \sim \text{Beta}(\alpha, \beta)$$
$$CVAP_t^{ACS} \sim \text{Binomial}(VAP_t^{ACS}, C_t)$$
$$CVAP_{t,b} \sim \text{Binomial}(VAP_b^{dec}, C_t)$$



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Posterior:

$$C_t \mid - \sim \text{Beta}(\alpha + CVAP_t^{ACS}, \beta + VAP_t^{ACS} - CVAP_t^{ACS})$$
$$CVAP_{t,b} \mid - \sim \text{BetaBin}(\, VAP_b^{dec}, \alpha + CVAP_t^{ACS}, \beta + VAP_t^{ACS} - CVAP_t^{ACS})$$

Block-Level Estimates

$$\text{BetaBinomial}(2, 82.9532, 8.31037)$$

$$\text{BetaBinomial}(1, 114.953, 0.310371)$$

$$\text{BetaBinomial}(3, 137.953, 344.31)$$

$$\text{BetaBinomial}(68, 193.953, 0.310371)$$

$$\text{BetaBinomial}(1, 205.953, 705.31)$$

$$\text{BetaBinomial}(12, 98.9532, 0.310371)$$

$$\text{BetaBinomial}(1, 88.9532, 23.3104)$$

$$\text{BetaBinomial}(3, 24.9532, 11.3104)$$

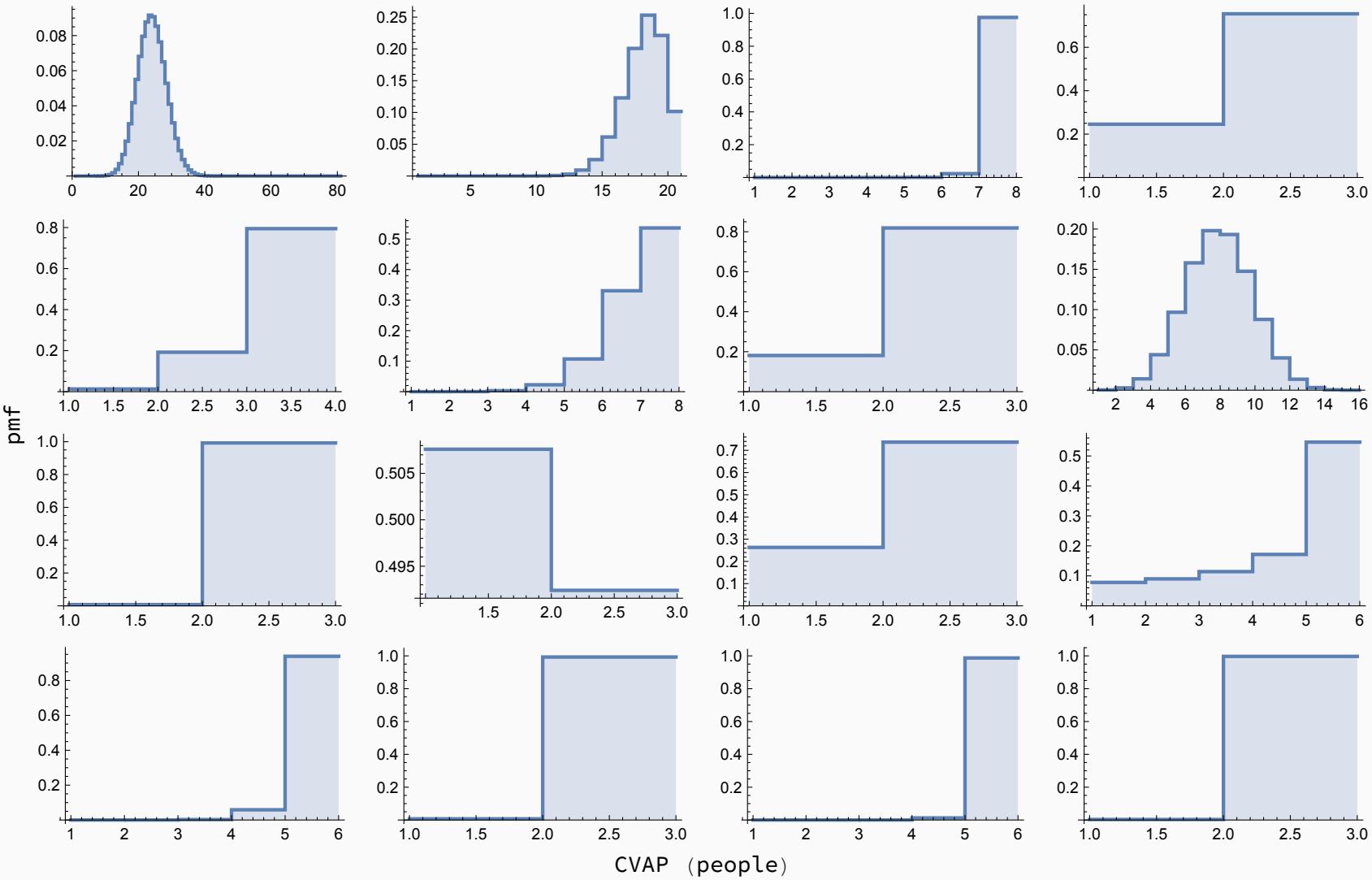
$$\text{BetaBinomial}(4, 191.953, 193.31)$$

$$\text{BetaBinomial}(1, 283.953, 90.3104)$$

$$\text{BetaBinomial}(1, 60.9532, 18.3104)$$

$$\text{BetaBinomial}(6, 46.9532, 16.3104)$$

Block-Level Estimates

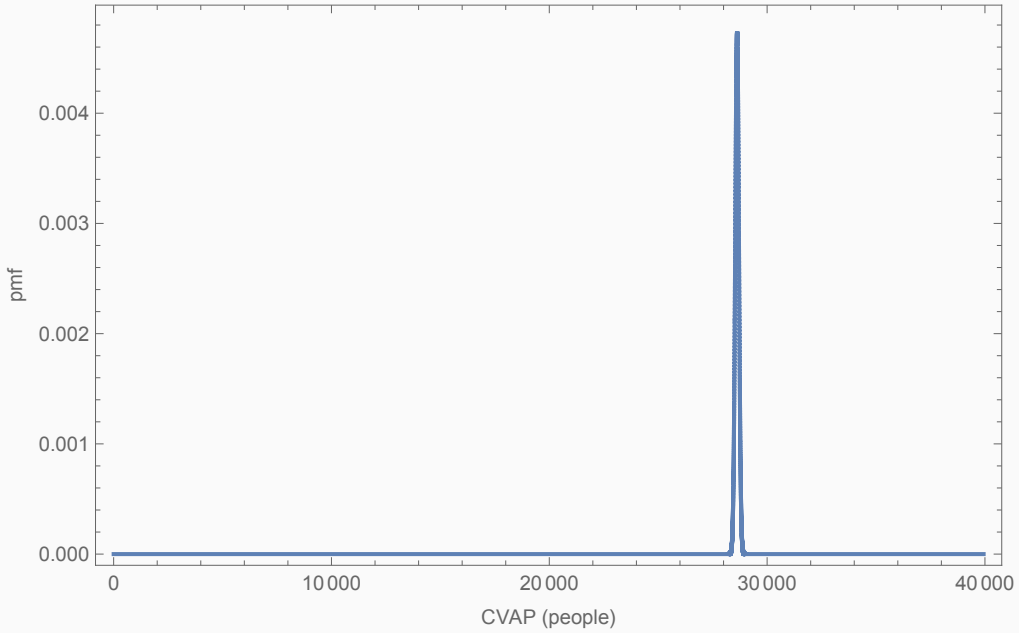


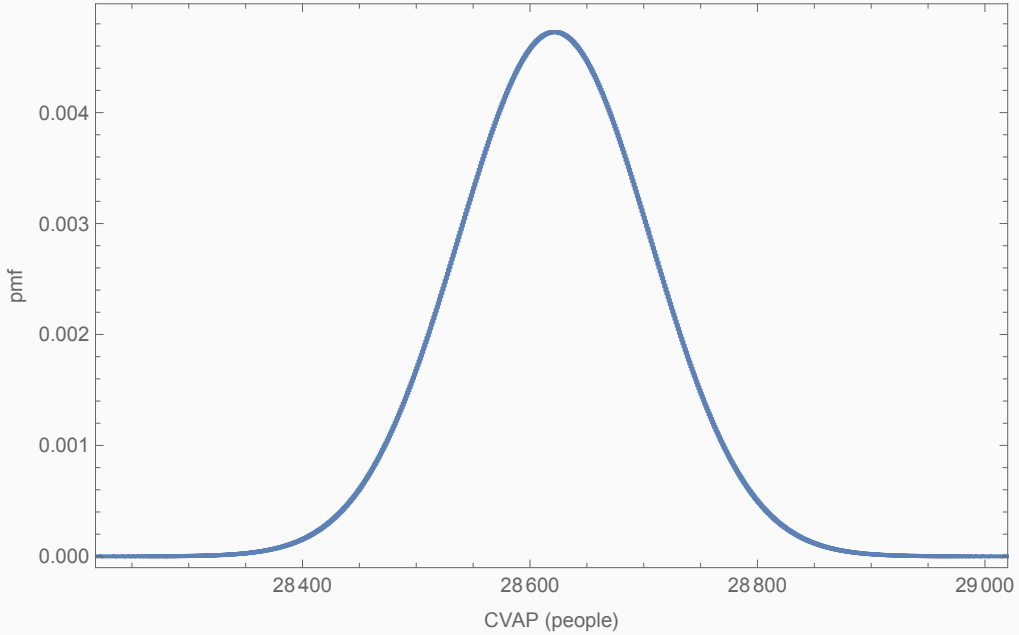
District-Level Estimates

Let d be a congressional district, $b_1, b_2, \ldots, b_n$ be the blocks in that district, $p * q$ be the convolution of p and q, and $\mathcal{F}$ be the Fourier transform.

$$CVAP_d = p_{b_1} * p_{b_2} * \cdots * p_{b_n} = \mathcal{F}^{-1}(\mathcal{F}(p_{b_1}) \cdot \mathcal{F}(p_{b_2}) \ldots \mathcal{F}(p_{b_n}))$$

District-Level Estimates





Race/Ethnicity Categories

Race/Ethnicity in the Census

➜ **NOTE: Please answer BOTH Question 5 about Hispanic origin and Question 6 about race. For this survey, Hispanic origins are not races.**

5 Is Person 1 of Hispanic, Latino, or Spanish origin?

- ☐ **No,** not of Hispanic, Latino, or Spanish origin
- ☐ Yes, Mexican, Mexican Am., Chicano
- ☐ Yes, Puerto Rican
- ☐ Yes, Cuban
- ☐ Yes, another Hispanic, Latino, or Spanish origin – *Print origin, for example, Argentinean, Colombian, Dominican, Nicaraguan, Salvadoran, Spaniard, and so on.*

6 What is Person 1's race? *Mark (X) one or more boxes.*

- ☐ White
- ☐ Black or African Am.
- ☐ American Indian or Alaska Native — *Print name of enrolled or principal tribe.*

- ☐ Asian Indian
- ☐ Chinese
- ☐ Filipino
- ☐ Other Asian – *Print race, for example, Hmong, Laotian, Thai, Pakistani, Cambodian, and so on.*
- ☐ Japanese
- ☐ Korean
- ☐ Vietnamese
- ☐ Native Hawaiian
- ☐ Guamanian or Chamorro
- ☐ Samoan
- ☐ Other Pacific Islander – *Print race, for example, Fijian, Tongan, and so on.*

- ☐ Some other race – *Print race.*

Race/Ethnicity in the Census

ACS race/ethnicity categories:

- White alone
- Black or African American Alone
- American Indian and Alaska Native Alone
- Asian Alone
- Native Hawaiian and Other Pacific Islander Alone
- Some Other Race Alone
- Two or More Races
- White Alone, Not Hispanic or Latino
- Hispanic or Latino

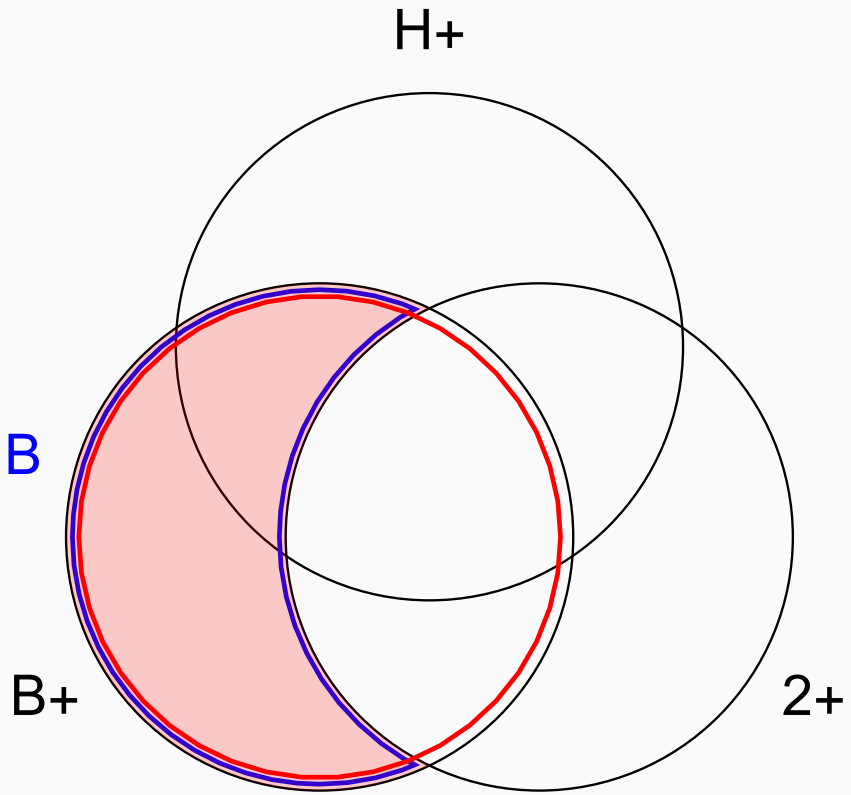
Litigation Race/Ethnicity Categories

Per Moon Duchin, the race/ethnicity categories relevant for litigation are:

- Any part African American or Black (including part Hispanic)
- Hispanic or Latino, but not Black
- Any part Asian or Pacific Islander, but not in any above category
- Any part Native American, but not in any above category
- Any part some other race, but not in any above category
- Any part white, but not in any above category

Venn Diagram

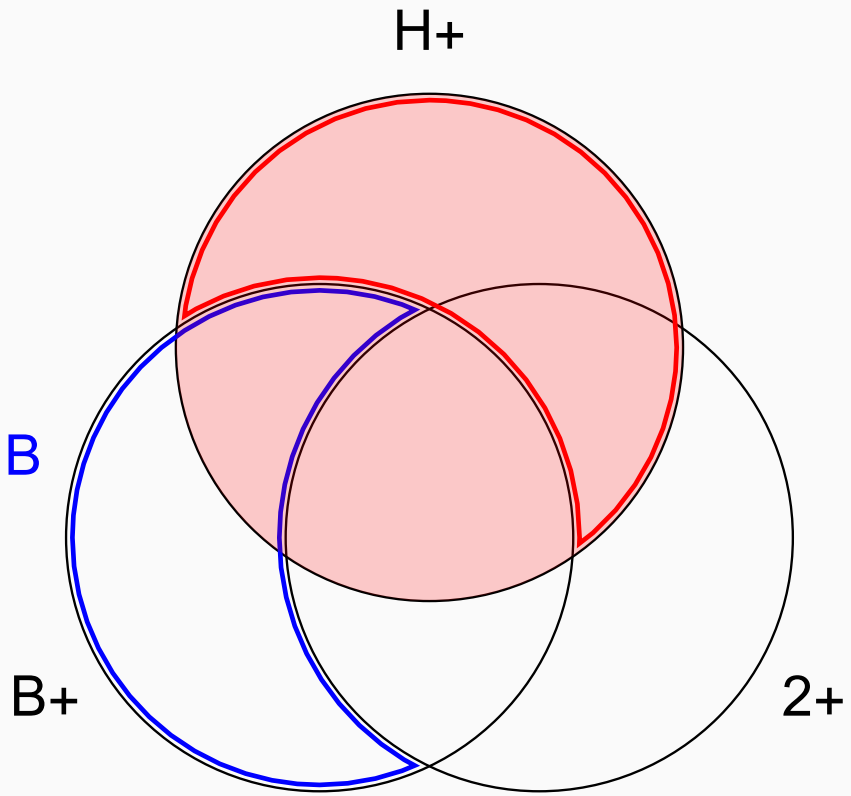
Any part African American or Black (including part Hispanic)



$$N(\text{B+}) \approx \underbrace{N(\text{B})}_{\text{ACS}} + \underbrace{N(\text{2+})}_{\text{ACS}} \underbrace{p(\text{B+} \mid \text{2+})}_{\text{Decennial}} = N(\text{B}) + N(\text{2+}) \frac{p(\text{B+}, \text{2+})}{p(\text{2+})}$$

Venn Diagram

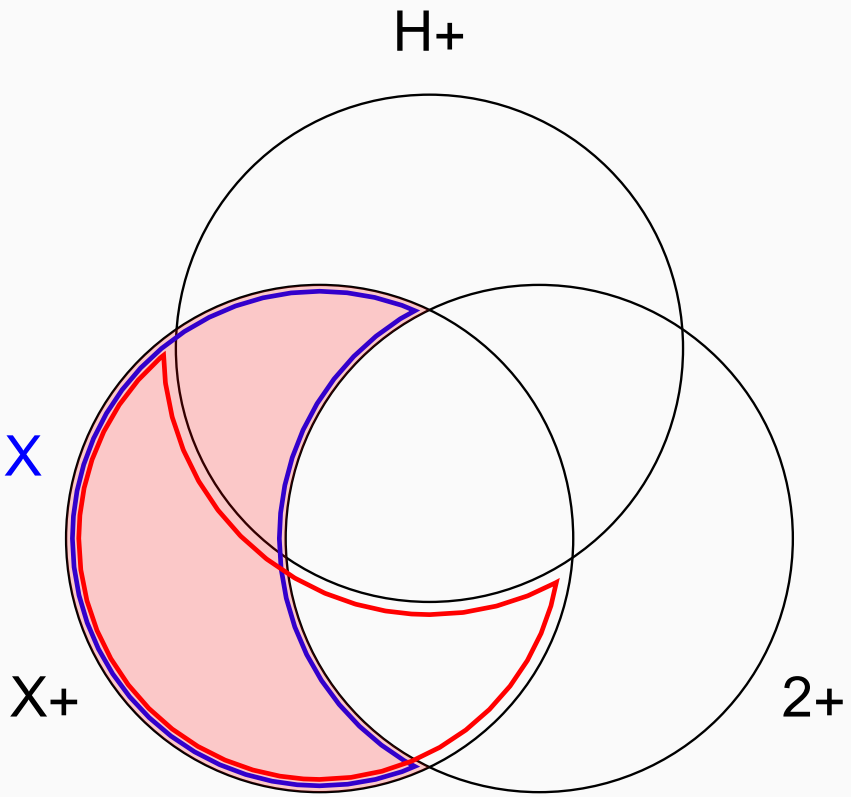
Hispanic or Latino, but not Black



$$N(\mathsf{H}+) - N(\mathsf{H}+)p(\mathsf{B}+ \mid \mathsf{H}+)$$

Venn Diagram

Any part X, but not in any above category



$$N(\mathsf{X}) - N(\mathsf{X})p(\mathsf{H+} \mid \mathsf{X}) + N(\mathsf{2+})p(\mathsf{T} \mid \mathsf{2+})$$

$$N(\mathsf{X}) - N(\mathsf{H+})p(\mathsf{X} \mid \mathsf{H+}) + N(\mathsf{2+})p(\mathsf{T} \mid \mathsf{2+})$$

Questions?